



Armada Kalamunda Group

Noradrenaline Infusion Guideline

1. Purpose

The purpose of this document is to guide staff caring for patients receiving Noradrenaline Infusions in the Critical Care environment to optimise and support cardiovascular function.

2. Applicability

This guideline applies to Medical, Nursing and Pharmacy staff working within the Intensive Care Unit at Armada Health Service.

3. Classification

Sympathomimetic

- Stimulates primarily alpha-adrenergic receptors to
 - Produce peripheral vasoconstriction (alpha adrenoreceptor action)
 - Produce a positive inotropic effect on the heart and coronary artery dilatation (beta adrenoreceptor action)

4. Indications

Hypotension secondary to septic shock, anaphylaxis (Adrenaline is the first line treatment), cardiogenic shock post myocardial infarction or after cardiac surgery.

Initiation of therapy should only occur after discussion with:

- ICU Consultant or Senior Registrar
- Emergency Department Consultant or Senior Registrar
- For theatre areas: The Consultant Anaesthetist (or after-hours the most senior Anaesthetist-on-site/Duty Anaesthetist)

5. Precautions

Infusion via Central Line / PICC line is preferred

- If central access is not available, consider use of a peripheral Metaraminol or Phenylephrine infusion
- Where peripheral Metaraminol or Phenylephrine is not suitable, a dilute Noradrenaline solution may be infused peripherally, through a large vein, until central access can be obtained.

- It must be determined that the peripheral cannula is flushing well and not tissue prior to initiating a peripheral Noradrenaline infusion.
- Blood Pressure parameters should be documented prior to **and on** commencement of Noradrenaline.
- Peripheral vasopressor therapy **is not a substitute** for correction of hypovolaemia, which should be assessed and corrected.
- Noradrenaline **CANNOT** be administered by IM, subcutaneous or IV bolus injection
- Prolonged use of sympathomimetics may result in a decrease of therapeutic effect (down-regulation of receptors). The patient should therefore be reassessed by medical staff regularly.
- Do not use if solution is discoloured or contains precipitate.

6. Adverse effects

- Common: transient headache, anxiety, arrhythmia
- Infrequent:
 - Hypertension

Inadvertent boluses may precipitate profound hypertension resulting in myocardial infarction and / or cerebral ischaemia / stroke syndrome.

- Bradycardia
- Extravasation of drug may cause peripheral ischaemia, sloughing, necrosis and gangrene.

Any suspected extravasation requires URGENT Care and medical review – See section 12 - Extravasation

- Rare: allergic type reactions including anaphylaxis (product contains sodium metabisulfite)

7. Contraindications

- Hypotension due to hypovolaemia
- Mesenteric or peripheral vascular thrombosis

8. Presentation

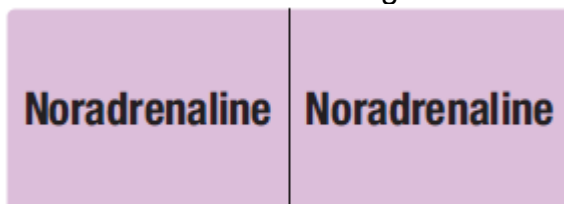
Noradrenaline (norepinephrine) acid tartrate monohydrate 4mg/4mL vial

9. Dosage / Preparation for Central Line

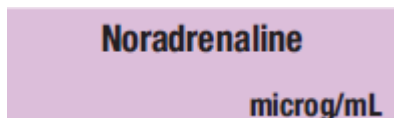
- Prescribe on Noradrenaline Intravenous Prescription Chart (AKMR176.2.3) to allow for titration of Noradrenaline to parameters specified by treating team.
- Usual starting rate: 3-5ml/hr, titrated to target blood pressure or mean arterial pressure (MAP) in steps of 1-2ml/hr (this may need to be done quickly in profound hypotension)
- The standard concentration is 8mg Noradrenaline in 100mL 5% glucose (if consultant prefers 0.9% Sodium Chloride, that is compatible as well)
- Patients in ICU may require large doses of Noradrenaline therefore more concentrated solutions of 16mg/100mL (**Double strength**) or 32mg/100mL (**Quad Strength**) may be used on the advice of an ICU physician.
 - 4mg Noradrenaline to 50mL via a syringe pump may be used in Theatre areas

- Preparation of the standard concentration of 8mg Noradrenaline in 100mL 5% glucose
 - The current Noradrenaline formulation is 4mg/4mls. Two vials are required.
 - **Remove 8mls** of fluid from the 100ml bag
 - Add the contents of the two vials of Noradrenaline (8mg = 8mls) to the bag
 - For higher concentrations; remove the equivalent volume from the 100ml fluid bag prior to adding the required dose (16mg = 16mls or 32mg = 32mls)
- Ensure the infusion bag is labelled on both sides.
 - The front label should be a square blue infusion label and must document:
 - Name of the medication
 - Dose, diluent and concentration.
 - Highlight the concentration if DOUBLE or QUAD strength.
 - Route i.e. CENTRAL
 - e.g. 8mg Noradrenaline in 100mL 5% glucose (CENTRAL) or 0.9% Sodium Chloride

- The back of the infusion bag **MUST** be labelled with a large Noradrenaline label



- Ensure line is labelled with small Noradrenaline sticker noting the concentration



- All dose titrations and weaning regimens should be aimed at the targeted Mean Arterial Pressure (MAP) which must be clearly documented by the medical team
 - When weaning the dose, it should be reduced gradually (avoid abrupt withdrawal)

10. Central line Administration (preferred route)

- Administer via B. Braun Infusion Pump as a continuous infusion. When medication libraries are activated, ensure the smart pump is set to 'Adult Critical Care' profile.

Use **ONLY** B.Braun Infusomat Space Line, SafeSet to administer Noradrenaline. These sets are **Orange and** have no side injection ports (to minimise the chance of inadvertent bolus injection).



- Do not gravity prime the line. Use the B. Braun pump to prime the line. This will prevent air introduction while priming the giving set.

10.1 Priming the Noradrenaline line using Braun Pump

- Pierce the new Noradrenaline bag with the giving set/ line and hang it to the IV pole.
- Press the drip chamber and fill it up to the green circle. And close the roller clamp.
- Ensure drip chamber is filled to the GREEN Line /Circle to avoid air entering the line. Air in the line will result in cessation of infusion to troubleshoot and will therefore create a drop in the patient's blood pressure.
- Turn the pump on, open the door, feed the giving set or line through the pump and close the door.
- Select the correct giving set for use.
- Then the pump will enter the Main Menu screen

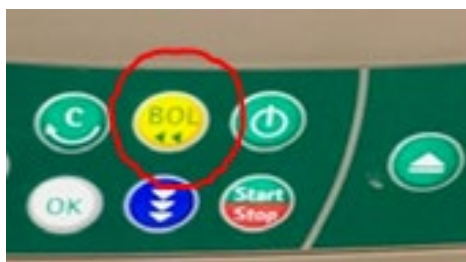


- Release the 'roller clamp' and remove the 'cap' from the end of the giving set/ line.
- Attach a 3 way tap to the end of the giving set / line.
- To minimise the chance of any accidental use of this port;
 - Attach a red bung to the unused port.
 - Attach a Noradrenaline sticker to the same port.

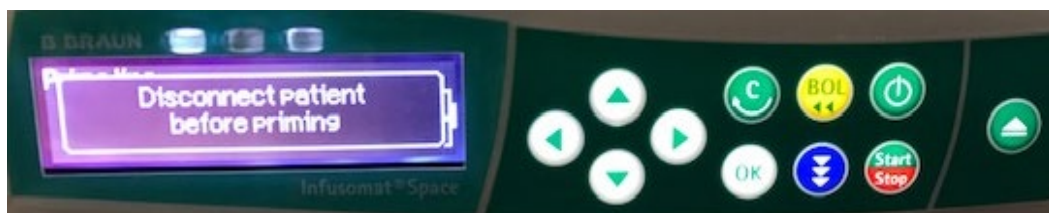


The unused, labelled port (with the red bung), is only to be utilised if a change in the Noradrenaline concentration is required or if there are no other lumen on the CVC line available to infuse other vasoactive medications, e.g. Vasopressin. The procedure for changing concentrations is detailed in section 10.2.

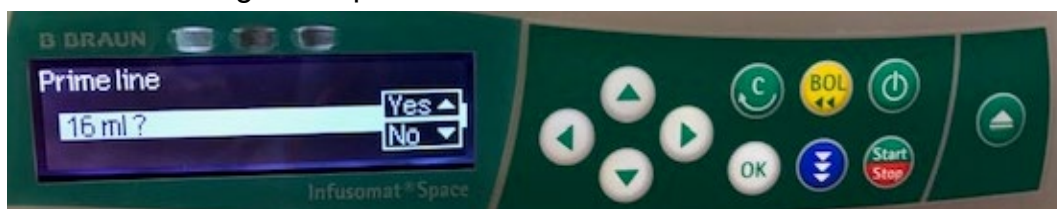
- Press the yellow 'Bolus key' to start priming the line.



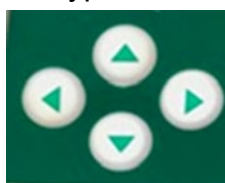
- A warning message will be displayed for few seconds to 'disconnect patient before priming' the line.



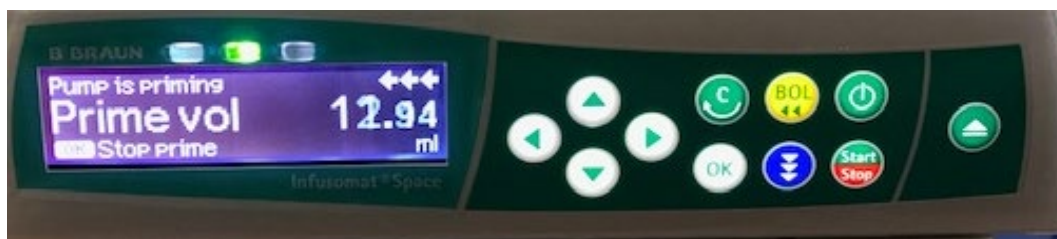
- Then the screen will change to a question 'Prime line 16 ml? Yes/No'



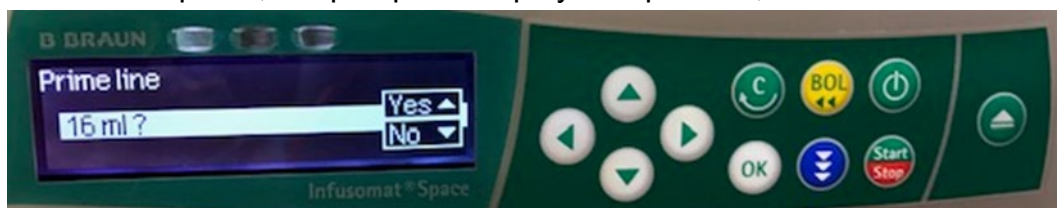
- Select 'Yes' using the multi directional keypad.



- The screen will change to a priming mode with rate 1200ml to quickly prime the line with 16 ml.



- This initial prime, will only prime $\frac{3}{4}$ of the administration set. To completely prime the line, we need to use the priming function again.
- After the initial 16mls prime, the pump will display the question, 'Prime line 16ml? Yes/No'.



- Select 'Yes' using the multi directional keypad to restart prime and fill the line completely.
- **Visually monitor that the line as it is primed. When the line is completely primed (to the end of the 3-way tap), stop priming by pressing the 'OK' key.**

- The pump will stop priming and will go back to 'Main menu'.



- Use the multidirectional keypad to enter the 'Rate' of the Noradrenaline infusion
- Press 'OK' Key to confirm
- Use the multidirectional keypad to go down, highlight and select the VTBI
- Enter VTBI of 70mls for the first bag.** Then press 'OK' Key again to confirm.

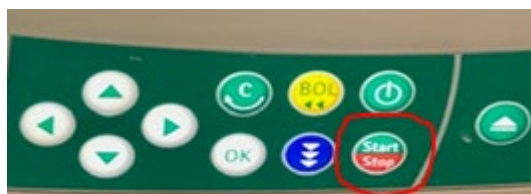
- Enter VTBI of 70mls for the Initial / first bag.**

- It takes approximately 10mls to fill the chamber up to the green circle and 20mls to prime the line.
- Filling the drip chamber to the green line and priming the administration set, uses a total of 30mls, leaving 70mls in the Noradrenaline bag.
- Setting the VTBI to 70mls is vital to get the warning alarm 10minutes prior to the end of the infusion.

If the VTBI is set incorrectly - the Noradrenaline bag may completely run out resulting in significant hypotension.

Enter 100mls as the VTBI for all subsequent bag changes

- Attach the giving set / line to CVC/ PICC line (blue lumen of the CVC preferred)
- Press 'start key' to start the infusion.



- Infuse through a dedicated line however if there is no other lumen available, Noradrenaline can be infused via a Y site with other compatible vasopressors / inotropes.

10.2 Cessation

Disconnect and discard infusion line if patient condition is stable and infusion is not actively in use after four (4) hours.

- When the Noradrenaline infusion is no longer required and has been ceased;
 - Disconnect the Noradrenaline line from the lumen, aspirate 5mls blood, then flush the lumen with 20 mls of Normal Saline to clear any Noradrenaline from the lumen using pulsatile method (push-pause)

- If unable to aspirate the lumen, attach a 10ml Syringe of Sodium Chloride 0.9% and infuse at a rate of 1- 0.5mls/hr to clear the lumen. If BP significantly increases, the rate needs to be decrease further or stop the infusion and put red cap and label "Do Not Use". If fail to aspirate, put red cap and label "Do Not Flush".

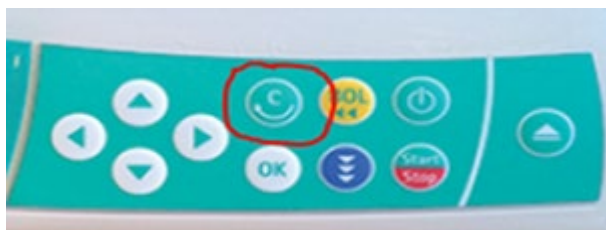
10.3 Noradrenaline infusion bag changing procedure

Noradrenaline infusion should not be paused or stopped. Stopping Noradrenaline (especially at a high rate) will cause a rapid decrease in Blood Pressure (BP).

- Changing the Noradrenaline bag whilst the infusion is running, must be done with extreme caution.
- The treating nurse must prepare a Noradrenaline infusion (same strength) two hours before the current infusion's finish time.

Remember **do not** press the 'start / stop key' at any stage while the infusion is running. This will stop the pump infusing Noradrenaline and can be dangerous.

- When the infusion bag has 10 minutes left to finish, the pump will give a warning alarm. It is important to set the volume to be infused (VTBI) at 70 ml after the first prime to ensure the pump alarms at the correct time.
- At the 10 minutes warning alarm, the Nurse should have the new Noradrenaline bag ready and press the 'C key'– (Main Menu/Cancel/Clear key) to access the main menu.



- When the C key is pressed, the pump will enter the Main Menu and the Rate will be highlighted.
- Use the 'down arrow' on the multidirectional keypad to highlight VTBI.



- Now select the VTBI by pressing the left 'arrow key' on the directional keypad.
- Change the VTBI to 100 ml (for all subsequent bag changes) using the directional keypad without delay. (If there is a delay the pump will automatically revert to the running mode **without** any change in the VTBI).
- After changing the VTBI to 100 ml, press 'OK key' to confirm the new VTBI.



- The pump will now display the new VTBI and then revert to the main page displaying continuous running indicator.



- If the 'OK' Key is not pressed, the pump will alarm to get your attention.
- Remove and discard the empty (old bag) of Noradrenaline from the giving set.
- Without delay, pierce the new Noradrenaline bag with the same giving set and hang it on the IV pole.
- Ensure no air enters the line and the drip chamber is filled to the green ring/line.
- The pump will keep running at the same rate, with the new VTBI and new Noradrenaline bag.

10.4 Change in Concentrations of Noradrenaline infusion (single to double, double to quad and vice versa) or Line change after 96 hours

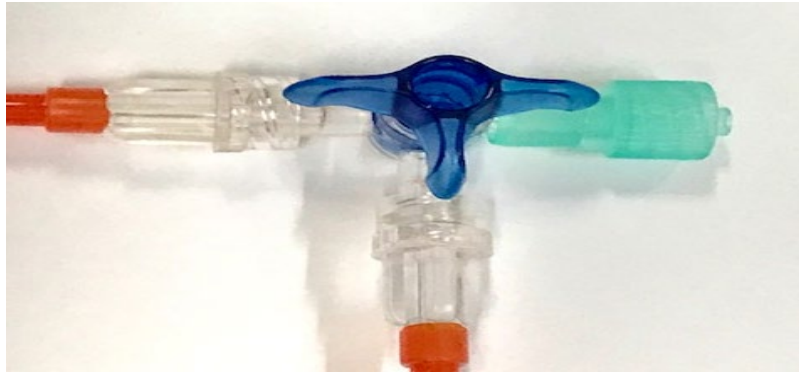
- Changing the Noradrenaline concentration from single to double or quad strength must to be done with extreme caution.
- Obtain a new Noradrenaline prescription on Noradrenaline Intravenous Prescription Chart (AKMR176.2.3) from the treating Medical staff.
- Prepare the new Noradrenaline bag with the new concentration (higher / lower) as previously described.
- Use a new Braun administration set (Ref 8700128SP - Orange colour) to prime the Noradrenaline infusion
- Prime the infusion line using the Braun pump as previously described.
- **Do not add a 3 way tap at the end of this Noradrenaline line.**
- When the line is completely primed, identify the current Noradrenaline infusion line. Locate the red bung at the end of the 3-way tap.



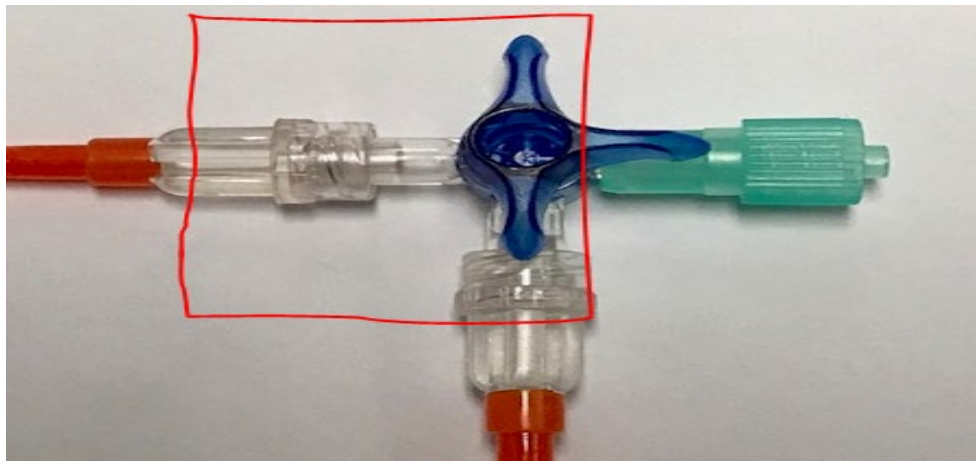
- Remove the red bung and attach this new line to the 3-way tap



- Turn the 3 way tap to enable both infusions to be run together as shown below.



- Closely monitor the patient's Mean Arterial Pressure (MAP).
- Start the new Noradrenaline pump at the appropriate rate (e.g.- half rate if the new infusion is double strength, double the rate if you are reducing concentrations)
- When the MAP goes up by 10mmHg, stop the first Noradrenaline infusion **and turn** the 3 way tap off to the first Noradrenaline infusion.



- Titrate Blood Pressure / MAP using the new infusion line and concentration.
- Remove the first Noradrenaline line and put a new red bung and the Noradrenaline sticker on the now unused port on the 3-way tap



- Observe the patient's MAP/BP for any changes and titrate infusion rate to target MAP.

11. Peripheral Noradrenaline administration

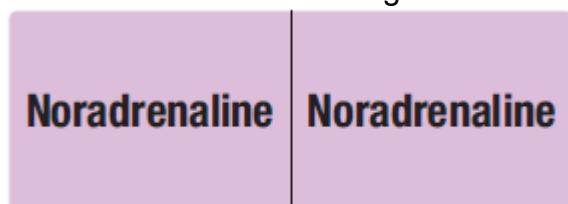
The use of Peripheral Noradrenaline is restricted to ED, Theatres, PACU and ICU. Only to be used in exceptional circumstances and only if Metaraminol infusion is not appropriate.

- Dilute 4mg to 500mL with compatible fluid
- Initiation of peripheral Noradrenaline should only be considered upon Consultant advice and should be reviewed regularly by the treating Consultant to monitor for ongoing requirement

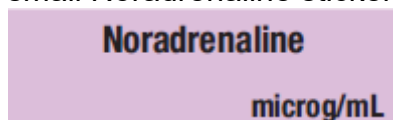
Ensure administration is via a large vein; in a visible site and contralateral to any blood pressure cuffs or tourniquets

- Avoid antecubital fossa IV line due to the increased risk of undetected extravasation or blockage/kinking.
- Usual starting rate: 20-40mL per hour) titrated to target blood pressure or mean arterial pressure specified by treating team
- Ensure the infusion bag is labelled on both sides.
 - The front label should be a square blue infusion label and must document:
 - Name of the medication
 - Dose: diluent and concentration - 4mg Noradrenaline in 500mL 5% glucose
 - Route: PERIPHERAL

- The back of the infusion bag **MUST** be labelled with a large Noradrenaline label



- Ensure line is labelled with small Noradrenaline sticker noting the concentration



Use **ONLY** B.Braun Infusomat Space Line, SafeSet to administer Noradrenaline. These sets are **Orange** and have no side injection ports (to minimise the chance of inadvertent bolus injection).



- Administer via B. Braun Infusion Pump (Smart Pump) as a continuous infusion.
- When medication libraries are activated, ensure the smart pump is set to 'Adult Critical Care' profile
- Alternatively, in theatre or PACU, administration may be via a syringe driver
- When peripheral Noradrenaline infusion has been ceased:
 - If peripheral venous access is no longer required, remove cannula
 - If peripheral access is still required, cease infusion and aspirate 5mls of blood from line. Flush slowly with 20mL of sodium chloride 0.9% or glucose 5% whilst monitoring patient's vital signs including blood pressure.
 - If unable to aspirate line,
 - Do not attempt to flush
 - Place a red cap/bung on the cannula
 - Place a Noradrenaline sticker on the line
 - Seek assistance from a senior medical officer

The duration of peripheral administration of Noradrenaline should be limited to 6 hours or less.

Longer durations represent a higher risk of extravasation and must be approved by a Consultant.

12. Monitoring

Continuous cardiac monitoring is essential.

Invasive haemodynamic monitoring via an arterial line is preferred and should be considered in all patients.

- Prescribe and manage according to the Noradrenaline Intravenous Prescription Chart (AKMR176.2.3).
- Targeted Blood pressure parameters must be clearly documented.
- Monitor infusion site for signs of extravasation or blanching of the vein.
- Visual checks of the infusion site should be performed every 30 minutes as the vasoconstrictive properties may cause local necrosis.
- Baseline and 4 hourly neurovascular observations should be done to assess risk of peripheral ischemia, when peripheral Noradrenaline infusions are used.
- An indwelling urinary catheter may be required for patients requiring hourly urine output monitoring
- Monitor for cardiac dysrhythmias and report any episodes to the medical team to clarify if a reduction in the rate is clinically indicated.

13. Extravasation

Seek immediate medical assistance if extravasation of peripherally or centrally administered Noradrenaline occurs

- Stop the Noradrenaline infusion immediately and aspirate any residual Noradrenaline from the IV line / cannula.
- Seek immediate senior medical or anaesthetic advice to initiate urgent treatment as listed below.
- The IV cannula should be removed when safe to do so. Ideally this should be done with a senior doctor present and alternative IV access provided
 - Liberally infiltrate the area as soon as practical with 5-10mg of Phentolamine in 10-15mL of sodium chloride 0.9% given as multiple injections across the affected area. This should be prescribed on the front of the WA Hospital Medication Chart (WA HMC)
 - Phentolamine is an SAS drug available in AKG ICU Fridge.

Caution: Phentolamine may cause hypotension due to systemic absorption

- Apply a 2.5cm strip of glyceryl trinitrate 0.2% ointment (Rectogesic®) to the affected area and cover this with Tegaderm® or apply a glyceryl trinitrate patch (5mg starting dose) to improve local blood flow. This can be re-dosed every 8 hours. This should be prescribed on the front of the WA HMC.

Caution: Glyceryl Trinitrate may cause hypotension due to systemic absorption

- Outline the extravasation with a surgical single patient marker to provide a baseline for monitoring.
- Consider the use of medical illustration photography for the medical record
- Consider elevating and warming the limb.
- Avoid cold packs or ice as this may worsen vasoconstriction

14. Compatible fluids

- Glucose 5% is preferred.

NB: Glucose 5% minimises oxidative degradation of Noradrenaline, but it may be harmful in high volumes for patients requiring resuscitation.

15. Drug interactions

- Do not mix Noradrenaline with alkaline solutions, iron salts or oxidising agents
- If sodium bicarbonate is also required to correct acidosis, administer via a separate line
- Use with caution in patients receiving monoamine oxidase inhibitors (MAOIs) or tricyclic antidepressants because severe and prolonged hypertension may result

16. Legislative context

The following documents are required to give effect to this policy:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 206
- Health Department of Western Australia (HDWA) [High Risk Medication Policy MP 0131/20](#)
- East Metropolitan Health Service (EMHS) [Consent Policy EMHS: 5](#)
- EMHS [Patient Identification and Procedure Matching EMHS: 21](#)

17. Supporting information

The following documents support the implementation of this policy:

- [Medication Policy AKG-POL-0253-13](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)

18. Compliance, monitoring and evaluation

Compliance against this guideline will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the department meeting for review and identification of any opportunities for improvement.

19. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3 Healthcare rights and informed consent
- The health service organisation uses a charter of rights that is:
 - a. Consistent with the Australian Charter of Healthcare Rights
 - b. Easily accessible for patients, carers, families and consumers
- Standard 4 – Action 4.3 Partnering with consumers
- Clinicians use organisational processes from the Partnering with Consumers Standard in medication management to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 4 – Action 4.15 High-risk medicines
- The health service organisation:
 - a. Identifies high-risk medicines used within the organisation
 - b. Has a system to store, prescribe, dispense and administer high-risk medicines safely
- Standard 5 – Action 6.5 Correct identification and procedure matching
- The health service organisation:
 - a. Defines approved identifiers for patients according to best-practice guidelines
 - b. Requires at least three approved identifiers on registration and admission; when care, medication, therapy and other services are provided; and when clinical handover, transfer or discharge documentation is generated

20. References

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21. Document contact

Enquiries relating to this guideline may be directed to:

Title: Nurse Unit Manager
Department: Intensive Care Unit

22. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3.0	12/03/2025	12/03/2028	Scheduled review

23. Authorisation

This guideline has been authorised and issued by the Director Clinical Services

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	12/03/2025
Published date	12/03/2025